Exponential generating functions and labelled structures

Notes from Carlos' lessons

Source lessons: 06-13 (EGFs and sums of powers), 06-21 (EGFs vs. algebraic datatypes), 07-07 ($e^{A(x)}$ and multinomial coefficients).

1 Definition: OGF vs. EGF (06-13)

	map	$(1, 1, 1, \dots) \mapsto$
ordinary (OGF)	$(a_0, a_1, \dots) \mapsto \sum a_n x^n$	$\dfrac{1}{1-x}$
exponential (EGF)	$(a_0, a_1, \dots) \mapsto \sum a_n \dfrac{x^n}{n!}$	e^x

(The board also recalls $e^x = \lim_{n\to\infty}(1 + \frac{x}{n})^n$ and $e = \lim(1 + \frac{1}{n})^n$.)

2 The product rule (06-21) and its meaning

Claim 1. *If* $A(x) = \sum a_n \frac{x^n}{n!}$ *and* $B(x) = \sum b_n \frac{x^n}{n!}$, *then*

$$A(x)B(x) = \sum_{n\geq 0} \Big(\sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k} \Big) \frac{x^n}{n!}.$$

Proof. $[x^n]AB = \sum_{k=0}^{n} \frac{a_k}{k!} \cdot \frac{b_{n-k}}{(n-k)!} = \frac{1}{n!} \sum_k \binom{n}{k} a_k b_{n-k}$. □

Why EGFs count labelled structures. Suppose a_n is the number of structures of type A on the label set $[n]$, and b_n likewise for B. A pair (an A-structure, a B-structure) on $[n]$ is built by: (1) choosing the size k of the label set of the left component; (2) choosing *which* k labels go left, in $\binom{n}{k}$ ways; (3) putting an A-structure on those labels and a B-structure on the remaining $n-k$, in $a_k b_{n-k}$ ways. That is exactly the coefficient above. (The board contrasts this with OGFs, where the product counts pairs with sizes adding, with no relabelling.) The sum $A + B$ counts "an A or a B structure" (the board's case classes $T_1 \cup T_2$), and $A \cdot B$ counts pairs $(T_1 \times T_2)$ with size(t_1, t_2) = size(t_1) + size(t_2).

Application: the binomial theorem (06-21 board)

Take $a_k = \alpha^k$ and $b_k = \beta^k$: $A(x) = \sum \frac{(\alpha x)^k}{k!} = e^{\alpha x}$ and $B(x) = e^{\beta x}$. Then $A(x)B(x) = e^{(\alpha+\beta)x}$ has n-th coefficient $(\alpha+\beta)^n$, while the product rule gives $\sum_k \binom{n}{k}\alpha^k\beta^{n-k}$. Hence

$$(\alpha+\beta)^n = \sum_k \binom{n}{k}\alpha^k\beta^{n-k}.$$

(Compare the combinatorial proof in the topic *Counting and double counting.*)

3 Sums of powers via EGFs (06-13)

Let $S_k(n) = \sum_{i=1}^n i^k$ for $k \geq 0$, so $S_0(n) = n$ and $S_1(n) = \frac{n(n+1)}{2}$.

Elementary (telescoping) derivation

$(i+1)^{k+1} - i^{k+1} = \sum_{\ell=0}^k \binom{k+1}{\ell} i^\ell$ by the binomial theorem. Sum over $i = 1, \ldots, n$; the left side telescopes:

$$(n+1)^{k+1} - 1 = \sum_{\ell=0}^{k} \binom{k+1}{\ell} S_\ell(n).$$

Isolating the $\ell = k$ term ($\binom{k+1}{k} = k+1$) gives the recursion boxed on the board:

$$S_k(n) = \frac{(n+1)^{k+1} - 1 - \sum_{\ell=0}^{k-1} \binom{k+1}{\ell} S_\ell(n)}{k+1}.$$

Example ($k = 2$). $(n+1)^3 - 1 = \binom{3}{0}n + \binom{3}{1}\frac{n(n+1)}{2} + 3S_2(n)$, i.e. $n^3 + 3n^2 + 3n = n + \frac{3}{2}n(n+1) + 3S_2$, so

$$S_2(n) = \tfrac{1}{3}n^3 + \tfrac{1}{2}n^2 + \tfrac{1}{6}n.$$

EGF derivation (the same identity, packaged)

Put $F_n(x) = \sum_{k\geq 0} S_k(n)\frac{x^k}{k!}$. Swapping the sums, $F_n(x) = \sum_{i=1}^n \sum_k \frac{(ix)^k}{k!} = \sum_{i=1}^n e^{ix} = \dfrac{e^{(n+1)x} - e^x}{e^x - 1}$ (a geometric series in e^x). The board writes the version summing from $m = 0$, $\frac{e^{(n+1)x}-1}{e^x-1}$, which differs only by the $i = 0$ term, i.e. it counts $0^0 = 1$ in S_0. Multiply by $e^x - 1 = \sum_{j\geq 1} \frac{x^j}{j!}$:

$$F_n(x)\,(e^x - 1) = e^{(n+1)x} - e^x.$$

Take $[x^{k+1}] \cdot (k+1)!$. The left side is the EGF product of $(S_\ell(n))_\ell$ with $(0, 1, 1, 1, \ldots)$, i.e. $\sum_{\ell=0}^k \binom{k+1}{\ell} S_\ell(n)$, and the right side is $(n+1)^{k+1} - 1$. This is the identity above. (It is the board's chain, rewritten with the EGF $e^x - 1$ of $(0, 1, 1, 1, \ldots)$.)

4 Algebraic datatypes as generating-function equations (06-21)

A recursive datatype defines a combinatorial class: sum types become +, product types become ×, and the size of a value is the sum of the sizes of its parts. Translate directly to an OGF equation (sizes only, no relabelling needed).

- **FibList.** `FibList = [] | 1 (FibList) | 2 (FibList)`, where the value 1 or 2 contributes size 1 or 2 (the board: "finite list consisting of 1s and 2s, size $= \ell_1 + \cdots + \ell_k$"). Then

$$F = 1 + xF + x^2F \;\Rightarrow\; F = \frac{1}{1 - x - x^2},$$

  so the number of lists of total size n is $[x^n]F = F_{n+1}$ (Fibonacci), equivalently the number of ways to climb n steps taking 1 or 2 at a time (the staircase pictures on the board, and the definition of F_n as the number of sequences $s_i \in \{1, 2\}$ with $s_1 + \cdots + s_k = n$). Compare the topic *Ordinary generating functions*.

- **BinaryTree.** `BinaryTree = Node Int | Tree BinaryTree Int BinaryTree`, with each `Int` of size 1:

$$B(x) = x + x\,B(x)^2.$$

  Write $B = x\,C(x^2)$ where $C = 1 + yC^2$ is the Catalan series ($y = x^2$). Indeed $x + x \cdot x^2C^2 = x(1 + x^2C^2) = xC$. So

$$B(x) = \frac{1 - \sqrt{1 - 4x^2}}{2x} = \sum_k C_k x^{2k+1}:$$

  the number of such trees with $2k + 1$ nodes is the Catalan number C_k. (The board's tree pictures show the size-5 case.)

5 $e^{A(x)}$: sets of structures (07-07)

Given an EGF $A(x) = \sum_{k\geq 1} a_k \frac{x^k}{k!}$ with $a_0 = 0$, the board studies $e^{A(x)} = \sum_{n\geq 0} \frac{A(x)^n}{n!}$ and extracts its coefficients with two rules:

1. $[x^i]A(x)^n = \sum_{i_1+\cdots+i_n=i} [x^{i_1}]A \cdots [x^{i_n}]A$ (expand the product),
2. $[x^i]C(x)D(x) = \sum_{s_1+s_2=i} [x^{s_1}]C\,[x^{s_2}]D$.

Therefore

$$[x^i]\,e^{A(x)} = \sum_{n\geq 0} \frac{1}{n!} \sum_{i_1+\cdots+i_n=i} \frac{a_{i_1} \cdots a_{i_n}}{i_1! \cdots i_n!}.$$

Combinatorial meaning. $A(x)^n$ counts ordered n-tuples of A-structures on a labelled set partitioned into n non-empty blocks. The blocks have *distinct* label sets, so reordering produces $n!$ different ordered tuples of the same unordered collection; dividing by $n!$ counts **sets** of n structures. So $e^{A(x)}$ is the EGF of "a set of A-structures" (with no restriction on how many).

The board's picture of the same principle:

- $A = (1, 1, 1, \dots)$ has EGF e^x ("sets": one structure on each label set), and $e^x - 1$ is "non-empty sets".

- Two applications are in the topic *Set partitions and involutions*: e^{e^x-1} (set partitions) and $e^{x+x^2/2}$ (involutions).

6 Binomial and multinomial coefficients (07-07, right side)

- $\binom{n}{k} = \#\{k\text{-subsets of } \{1,\dots,n\}\} = \binom{n}{k,\,n-k}$ ("A and its complement").
- **Multinomial.** $\binom{n}{k_1,\dots,k_p}$ (with $k_1+\cdots+k_p=n$) counts the ordered partitions of $\{1,\dots,n\}$ into blocks of sizes $k_1,\dots,k_p$.

Proof. Choose the first block, then the second from what remains, and so on:

$$\binom{n}{k_1}\binom{n-k_1}{k_2}\binom{n-k_1-k_2}{k_3}\cdots\binom{k_p}{k_p} = \frac{n!}{k_1!(n-k_1)!}\cdot\frac{(n-k_1)!}{k_2!(n-k_1-k_2)!}\cdots = \frac{n!}{k_1!\cdots k_p!},$$

because the intermediate factorials cancel (the slash marks on the board) and $\binom{k_p}{k_p}=1$. □

Notes

- The right half of the 06-13 board also lists $\sum_n \frac{e^{(n-1)x}-1}{e^x-1}$ as an alternative form of $F_n(x)$ (consistent up to an index shift) and the product/EGF rewriting $E(a_k)\cdot E(b_k) = \sum \frac{x^n}{n!}\sum_k \binom{n}{k} a_k b_{n-k}$.
- The large grid sketch at the upper left of the 06-21 board (curves and a staircase between n and h) is a staircase/lattice picture that I could not interpret with confidence; I have only transcribed the algebra beside it.